

Hopf bifurcation

Motivating example

Lorenz eqns

$$\left. \begin{aligned} \dot{x} &= \sigma(y-x) \\ \dot{y} &= rx - xz - y \\ \dot{z} &= xy - bz \end{aligned} \right\} \quad \begin{aligned} r &= \text{bifurcation parameter} \\ \sigma, b &> 0 \end{aligned}$$

Reflection symmetry

$$\begin{aligned} \dot{X} &= V(X) \\ V(\gamma X) &= \gamma V(X) \\ \gamma^2 &= Id \end{aligned}$$

$$\gamma = \begin{bmatrix} -1 & & \\ & -1 & \\ & & 1 \end{bmatrix}$$

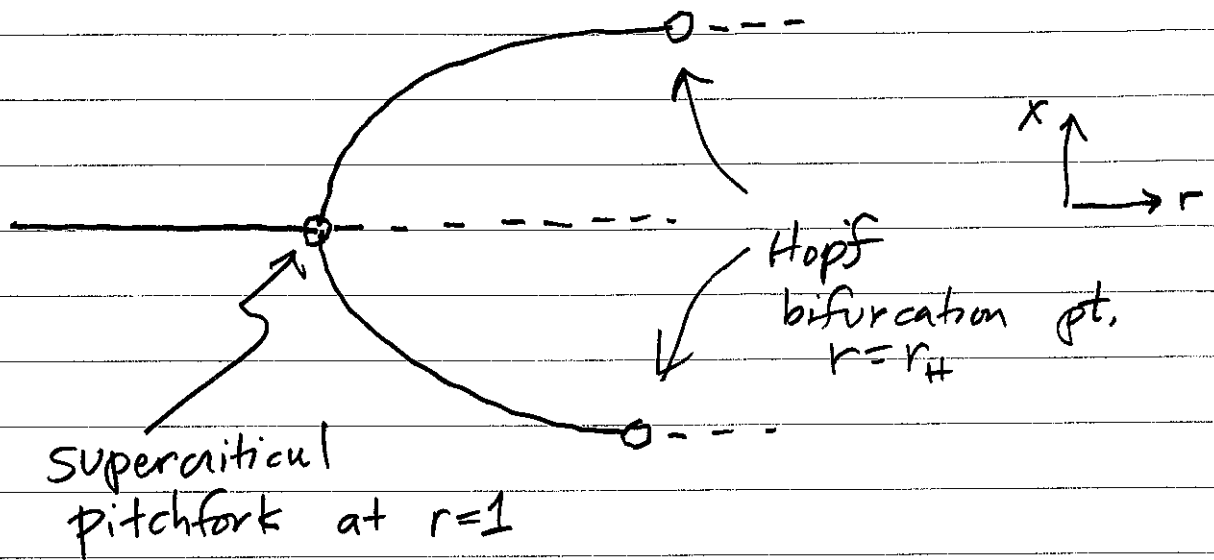
Trivial equilibrium $X^* = (0,0,0)$ is reflection symmetric
 $\gamma X^* = X^*$

other equilibria?

$$\begin{aligned} x &= y \\ z &= x^2/b \\ (r - 1 - x^2/b)x &= 0 \\ \Rightarrow x &= \pm \sqrt{b(r-1)}, \quad r > 1 \end{aligned}$$

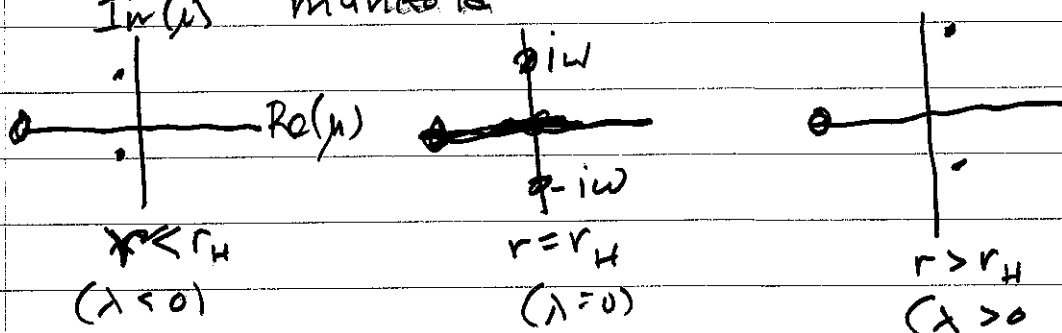
$$X_{\pm} = (\pm \sqrt{b(r-1)}, \pm \sqrt{b(r-1)}, r-1)$$

$\gamma X_+ = X_-$, $\gamma X_- = X_+$ & equilibria related by symmetry.

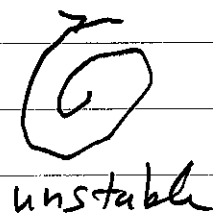
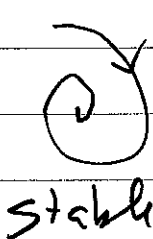


What can we say about behavior near the Hopf bifurcation?

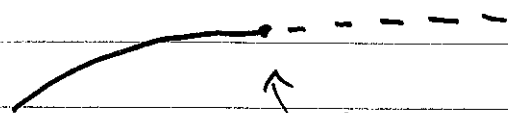
$\lambda_0 = (r = r_H)$
 $D_x V(x_{\pm}; \lambda_0)$ has a purely imaginary pair of eigenvalues $\mu_{\pm} = \pm i\omega$ and one eigenvalue $\mu_0 < 0$. (1-d stable manifold + 2-d center manifold)



Linear problem on 2-d invariant manifold

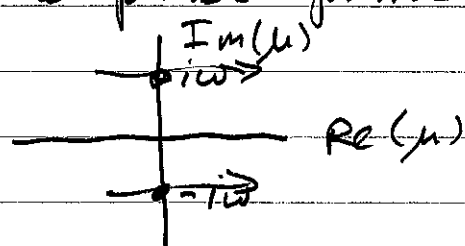


Since $\text{Det}[DV] \neq 0$, then no loss (or gain) in equilibria^x at Hopf bifurcation



can we say anything generic about what happens here?
Yes! Hopf bifurcation then

Simple example of Hopf bifurcation in the phase plane



$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \lambda & -\omega \\ \omega & \lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} (x^2+y^2)x \\ (x^2+y^2)y \end{pmatrix}$$

$$\begin{aligned} \dot{x} &= \lambda x + \overset{-\omega y}{a(x^2+y^2)x} - b(x^2+y^2)y \\ \dot{y} &= \omega x + \lambda y + b(x^2+y^2)x + a(x^2+y^2)y \end{aligned}$$

Equations are rotationally symmetric

If $\begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$ is a soln then so is

$$R_\theta \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \underbrace{\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}}_{R_\theta} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

$$\dot{X} = V(X) \quad \text{and} \quad R_\theta V(X) = V(R_\theta X) \quad \text{for all } \theta$$

$$\Rightarrow R_\theta \dot{X} = R_\theta V(X) = V(R_\theta X) \quad \text{follows since}$$

$$\begin{aligned} \text{i.e. } \dot{X} = V(X) \\ \Rightarrow (R_\theta X)' = V(R_\theta X) \end{aligned} \quad \left\{ \begin{array}{l} R_\theta \begin{pmatrix} x & -\beta \\ \beta & \alpha \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} R_\theta \\ \& (x^2 + y^2) \text{ is rotationally symmetric} \end{array} \right.$$

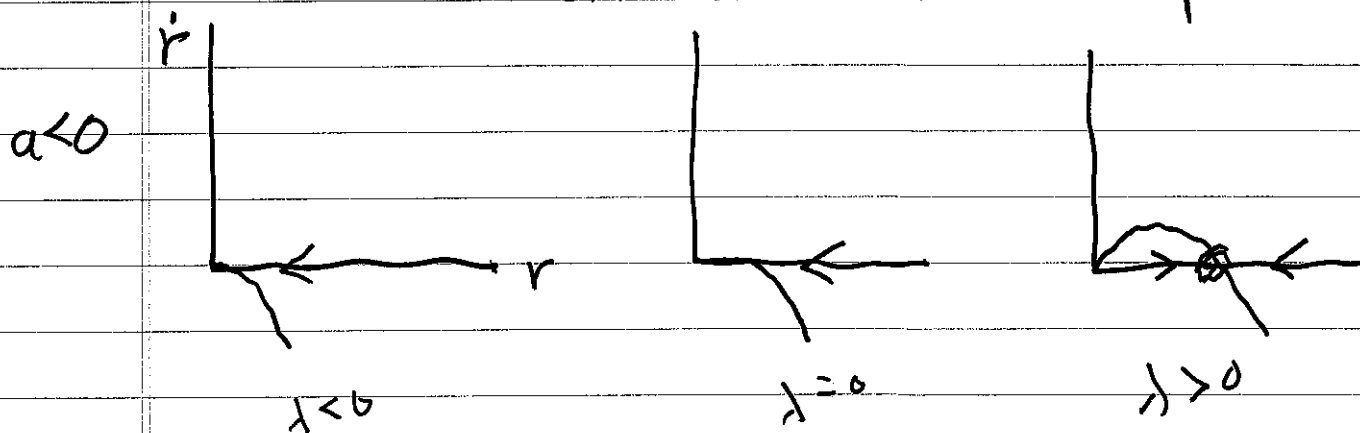
Consequence of rotational symmetry is that the eqns. take a very simple form in polar coordinates

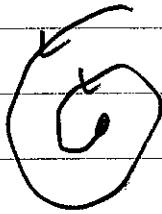
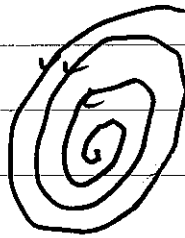
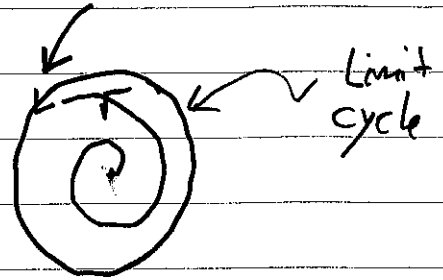
$$\begin{pmatrix} \dot{r} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} f(r) \\ g(r) \end{pmatrix} = \begin{pmatrix} \lambda r + a r^3 \\ \omega + b r^2 \end{pmatrix}$$

RHS doesn't depend on θ

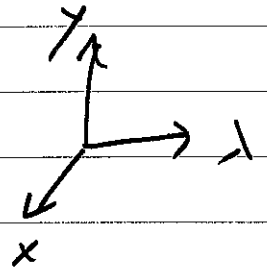
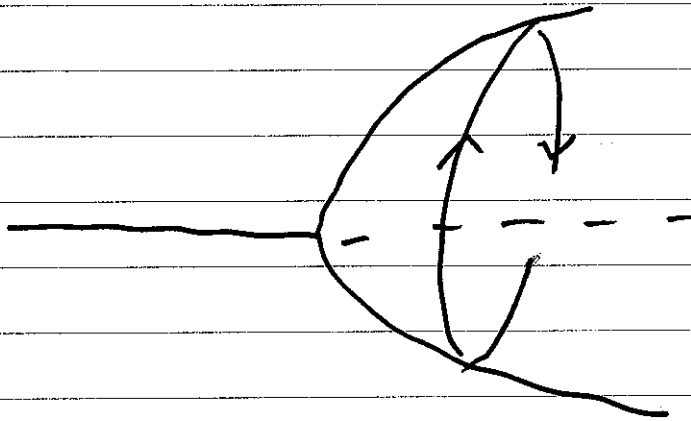
$\dot{r}$ -eqn is like pitchfork bif.

$a > 0 \Rightarrow$ subcritical & $a < 0 \Rightarrow$ supercritical



Phase Plane: $\lambda < 0$  $\lambda = 0$  $\lambda > 0$

Limit cycle



Hopf bifurcation produces a limit cycle in CM with small amplitude ($\sim \sqrt{\lambda}$ in size) with angular frequency ($\sim \omega$) in neighborhood of bifurcation pt ($x=0, \lambda=0$). If supercritical it exists for $\lambda \geq 0$, when fixed-pt. is unstable. If subcritical, then it exists for $\lambda < 0$, when fixed-pt. is stable.

More generally, after center manifold reduction we could assume that at the Hopf bifurcation

$$\begin{aligned}\dot{x} &= -\omega y + p(x, y) \\ \dot{y} &= \omega x + q(x, y)\end{aligned}$$

, p, q are $O(x, y)$



Lorenz:

$$(X, Y, Z)$$

at Hopf:

$$\begin{aligned}\dot{x} &= -\omega y + F(x, y, z) \\ \dot{y} &= \omega x + G(x, y, z) \\ \dot{z} &= -\lambda_0 z + H(x, y, z)\end{aligned}$$

center-manifold:

$$z = h(x, y)$$

$$\Rightarrow \begin{cases} p(x, y) = F(x, y, h(x, y)) \\ q(x, y) = G(x, y, h(x, y)) \end{cases}$$

where p, q have Taylor expansions:

$$\left. \begin{aligned}p &= A_1 x^2 + B_1 x y + C_1 y^2 + D_1 x^3 + \dots \\ q &= A_2 x^2 + B_2 x y + C_2 y^2 + D_2 x^3 + \dots\end{aligned} \right\} \begin{array}{l} \text{not} \\ \text{rotationally} \\ \text{symmetric} \\ \text{;} \end{array}$$

But we can make a "near-identity nonlinear coordinate transformation" to put this into "normal form", which is problem we just did.

Normal Form Transformation

$$\dot{X} = \underbrace{AX}_{\text{linear}} + \underbrace{g_2(X)}_{\text{quadratic}} + \underbrace{g_3(X)}_{\text{cubic}} + \underbrace{\mathcal{O}(X^4)}_{\text{remainder}}$$

$$X = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$AX = L = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix} \quad (\omega=1)$$

$$\text{let } X = Y + \underbrace{P_2(Y)}_{\mathcal{O}(Y^2)}$$

this is first
in a sequence of
"near-identity" coord.
transformations

Choose $P_2(Y)$ so that $\dot{Y}$ is "simpler" than $\dot{X}$ eqn.

After we do this, we let $Y = Z + P_3(Z)$
& choose $P_3(Z)$ so that $\dot{Z}$ eqn. is
simpler than $\dot{Y}$ and so on.
(actually, we stop there)

We can choose $P_2(Y)$ & $P_3(Z)$ so that

$$\begin{pmatrix} \dot{z}_1 \\ \dot{z}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} + \begin{pmatrix} a(z_1^2 + z_2^2) & -b(z_1^3 + z_2^3) \\ b(z_1^3 + z_2^3) & a(z_1^2 + z_2^2) \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}}_{\text{Rotationally symmetric}} + \mathcal{O}(z^4)$$